

MARTY SCOTT

(they/them)

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A software engineer and game designer with a CS degree and an MFA from NYU. I build game engines, gameplay systems, and tools from the ground up - from a 100k+ line Vulkan engine with ray tracing and accessibility features, to narrative state machines and physics-driven interactions in Unity and Unreal. I care about code that feels good to use, looks right on screen, and ships.

SKILLS

Languages: C#, C++, Java, Python, JavaScript, SQL, Custom Scripting

Engines / Tools: Unity, Unreal, GameMaker, TEGE (open source, BSL 1.1)

Engine Development: C++20, Vulkan, ECS architecture, visual scripting, shader authoring

Gameplay Systems: Encounters, interactive objects, narrative scripting, state machines, behavior trees, combat systems, 3D math

Web Development: Next.js, React, HTML/CSS, SEO

Creative Coding: p5.js, Processing

Design Tools: Blender, Photoshop

Audio Integration: FMOD, Pro Tools, Logic, Ableton

EXPERIENCE

NYU Game Center (Tisch) - Code Help Desk | 2025 - Present

- Help BFA and MFA students debug and fix projects across Unity, Unreal, Processing, and GameMaker
- Troubleshoot broken workflows and translate fixes into clear next steps

NYU Game Center - Studio Game Director & Producer | Spring 2025

- Led day-to-day production for a multidisciplinary game team from prototype through milestone builds
- Coordinated playtests, translated findings into prioritized next steps, kept scope aligned with deadlines
- Managed build stability, presentation flow, and last-mile polish for demo readiness

Custom Construction Website - Designer & Programmer | 2024

- Built a company website with editable content tools and improved SEO

PROJECTS

TEGE - The Enjin Game Engine | Solo Developer | C++20, Vulkan

- Architected and shipped a 100k+ line aesthetic-first game engine with a complete ImGui editor, 70+ ECS components, and build pipeline
- Implemented a Vulkan PBR renderer with cascaded shadow maps, clustered forward lighting, GPU occlusion culling, and full hardware ray tracing
- Integrated dual physics backends (Jolt 5.2.0, Box2D 3.0.0), AngelScript scripting (721 bindings), and blueprint-style visual scripting with 146+ nodes
- Built a 30+ effect post-processing stack, retro rendering modes (PSX, Dreamcast), and 44 starter templates
- Developed accessibility suite: 8 colorblind corrections, remappable input, screen reader, switch access, dwell-click, dyslexia mode
- Published under BSL 1.1 with full documentation, installer, and standalone game player

RhodesTrip | Producer / Designer / Programmer / Audio | Unity, FMOD, Ink, Blender

- Built narrative state machine, cameras, vehicle controls for a third-person driving + exploration game
- Owned prototype slice, iteration based on playtests, cross-discipline coordination (team of 4)

IRIS (In-Progress) | Solo Developer | Unity

- Third-person environmental exploration with physics-based flower trimming and arranging
- Prototyped tactile interactions; iterated on feel and reliability

Blackout | Solo Developer | Unity, FMOD

- Sightless 2D platformer: audio-driven feedback, navigation cues, mix logic, accessibility hooks, debug tools
- Published research whitepaper on designing for blind play

Cold Flame | Solo Developer | Unity

- Autobiographical physics game; 9-minute experience using physics mechanics as emotional metaphor

Purrfect Pizza VR | Design / Music / Audio Programming (Team of 6) | Unity, VR Toolkit, Logic

- VR game gamifying formal language constructs through pizza-making for cat bosses

Prototype Studio Gallery | 11 Weekly Prototypes | Unity

- One per week; text adventures, physics toys, 3D environments, naval strategy, abstract interactions

EDUCATION

NYU (Tisch) - MFA, Game Design (Game Center) | 2024 - 2026

Cal Poly Pomona - BS, Computer Science | 2021 - 2023

Mt. San Antonio College - AA/AS, Database Management & Programming | 2018 - 2021

Musicians Institute - AA, Piano Tech (Audio Engineering emphasis) | 2009 - 2011